Final exam study guide

How to prepare — cumulative (Units I–III)

Last updated: October 12, 2026.

Exam: Final exam · Wednesday, December 16, 10:20 a.m.–12:10 p.m., FR 201. Practice problems: Practice packet. Schedule: Schedule.

Recommended study plan

1. **Scan the learning outcomes below** (and the linked midterm study guides). Mark each as: confident / shaky / blank.
2. **Rework Unit I and Unit II with the midterm packets.** Use Midterm 1 study guide / practice and Midterm 2 study guide / practice for fluency you have already practiced once.
3. **Go through Lectures 35–40** for material since Midterm 2 (eigenvectors, complex numbers, least squares, SVD concepts). Match shaky outcomes to fill-ins, worked examples, and Check yourself items.
4. **Build a cumulative "important definitions and formulas" list.** Include vectors and geometry, matrices / Ax / systems, dynamics, eigenpairs, least-squares residual language, and SVD rank-k ideas — not only the newest topics.
5. **Rework quiz styles** from the whole term (Quizzes 1–11). Cover solutions, redo by hand, then check filled-in notes in Canvas.
6. **Start memorizing your list.**
7. **Try the final practice packet with no notes** — Sections A–C by unit, then Section D mixed.
8. **Check answers** (end of the practice packet). For every miss, update your list and redo a similar lecture or midterm-practice item.
9. **Memorize the final list** so you can write it from memory before the exam. In class you will not have notes.

Useful course materials

Source	Role
Lectures 01–40	Full course (skip midterm / Veterans Day / fall-break days as noted on the schedule)
Midterm 1 study guide + practice	Unit I refresh (Lectures 01–14)
Midterm 2 study guide + practice	Unit II refresh (Lectures 18–29)
Quizzes 1–11	Fluency checkpoints across the term
Final practice packet	Cumulative practice (Units I–III + mixed)
Boyd & Vandenberghe, *VMLS*	Textbook (free online)
Strang §6.1	Eigenvalues / characteristic equation
Lecture 33	SVD and low-rank context (conceptual fluency on the final)

Unit I — Vectors and geometry (Lectures 01–14)

Use the Midterm 1 study guide for the full outcome list. High-priority skills for the final:

1. Represent data as vectors; add, scale, and form linear combinations.
2. Compute and interpret inner products, norms, distances, RMS, angles, and cosine similarity.
3. Distinguish linear vs affine models; evaluate $a^\mathsf{T}x + b$ in context.
4. Run a small nearest-neighbor or k-means step by hand and discuss scale.
5. Use dependence, span, subspace, basis, dimension, and orthonormal language accurately.

Unit II — Matrices, systems, and dynamics (Lectures 18–29)

Use the Midterm 2 study guide for the full outcome list. High-priority skills for the final:

1. Compute Ax, interpret columns, and recognize common 2D transforms.
2. Translate contexts into $Ax = b$; solve small systems; read RREF (pivots, free variables, consistency).
3. Model $x_{t+1} = Ax_t$; compute a few steps; relate $A^k x$ to long-run behavior.
4. Multiply matrices; compose transforms; compare computational orders.
5. Use invertibility, rank, column space, null space, and 2×2 / triangular determinants.